

Kent Events

Testing Report

COMP6030

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Computing

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Introduction

This testing report provides supporting evidence for the testing and validation claims made in the main technical report. It focuses on backend API testing because the main system risk was contract correctness across authentication, event management, database state, and frontend/API integration. The tables show how test coverage evolved across development iterations, and how failing behaviours were identified and corrected.

1. Testing Model, Environment, and Isolation

The backend was tested using an end-to-end style approach, where tests made HTTP requests against the API in a similar way to the frontend client. Rather than only testing isolated service functions, the tests validated complete route behaviour, including request validation, status codes, response structures, database side effects, session behaviour, and external side effects such as verification email generation. This made the tests useful for checking whether the API behaved correctly from the perspective of a real client.

The authentication tests covered registration, login, logout, email verification, resend verification, and authenticated user retrieval through /me. These tests were especially important because the authentication model evolved from a simpler flow into a session-based system with persistent login state and email verification. The events tests covered creation, retrieval, updating, deletion, filtering, ownership checks, image retrieval, and image update behaviour. Together, these suites provided regression coverage as both the event model and authentication model changed during development.

A practical limitation of the test environment was that the same database had to be used for both the live web application and backend testing. This meant the tests could not simply truncate tables before or after each run, as doing so could remove real application data. Instead, each test created users and events with unique generated identifiers, such as prefixed emails or event titles. These prefixes allowed test-created records to be found reliably and removed during cleanup without affecting unrelated data. Although this was less convenient than using a fully isolated test database, it forced the test suite to be more precise about which records it created, used, and deleted.

Test isolation was therefore achieved through unique test data generation, targeted database cleanup, and the resetting of mocked external dependencies such as the verification email inbox. This helped ensure that each test could run deterministically without relying on the database being empty, while still preserving existing non-test data.

2. Test Suite Coverage by Iteration

The behaviours selected for testing were chosen to cover the main API contract categories: successful requests, invalid input, unauthenticated access, unauthorised access, missing resources, empty result sets, and image-specific edge cases. This made the tests useful for detecting regressions in frontend/backend integration as the event DTO, authentication model, and filtering behaviour changed during development.

The final automated backend suite consisted of **49 passing Vitest tests** across the authentication and event routes: **16 authentication tests** and **33 event tests**.

2.1. Authentication Test Suite

✓ passed, ✗ failed/exposed an issue, - not yet tested, and ▶ skipped

Route	Behaviour	Iter 1	Iter 2	Iter 3	Iter 4	Notes
POST /register	201 creates student user and sends email	✓	✓	✓	✓	Iter 4 replaced token link with 6-digit code
	201 creates society user and sends email	-	-	-	✓	Added with account types
	400 invalid/missing fields	✗	✓	✓	✓	Missing validation
	409 duplicate email	✓	✓	✓	✓	
GET /verify	200 valid token verifies user	✓	✓	✓	-	Replaced in Iter 4 by POST /verify
	400 invalid/expired token	✓	✓	✓	-	Replaced in Iter 4 by POST /verify
POST /verify	200 valid code verifies user	-	-	-	✓	Added with new email verification flow
	400 invalid/expired code	-	-	-	✓	Added with new email verification flow
POST /login	200 valid credentials	✓	✓	✓	✓	
	400 already logged in	-	-	✓	✓	Added with session persistence
	401 invalid credentials	✓	✓	✓	✓	
	403 unverified email	✓	✓	✓	✓	
POST /request-verify	200 send email (existing user)	✓	✓	✓	✓	
	200 no email (nonexistent user)	✓	✓	✓	✓	
POST /logout	200 clears session	-	-	✓	✓	Added with session persistence
	200 when not authenticated	-	-	✓	✓	Added with session persistence
GET /me	200 returns user	-	-	✓	✓	Added with session persistence
	401 not authenticated	-	-	✓	✓	Added with session persistence

2.2. Events Test Suite

✓ passed, ✗ failed/exposed an issue, - not yet tested, and ▶ skipped

Route	Behaviour	Iter 1	Iter 2	Iter 3	Iter 4	Notes
POST /create-event	201 creates event (full data)	✓	✓	✓	✓	
	201 creates event (no image)	✓	✓	✓	✓	
	400 invalid/missing fields	✓	✓	✓	✓	
	401 unauthenticated	✓	✓	✓	✓	
PUT /update-event	200 updates event details	✓	✓	✓	✓	
	200 preserves image when new one isn't sent	✓	✓	✓	✓	
	200 replaces image when new one is sent	✓	✓	✓	✓	
	200 clears image when <i>remove_image</i> is sent	-	✓	✓	✓	Added with improved event image handling
	400 when image and <i>remove_image</i> are sent	-	✓	✓	✓	Added with improved event image handling
	400 missing/invalid <i>eventId</i>	✗	✓	✓	✓	Missing validation
	401 unauthenticated	✗	✓	✓	✓	Missing auth check
	403 unauthorised update by a different user	✗	✓	✓	✓	Missing ownership check
	404 event not found	✓	✓	✓	✓	
DELETE /delete-event	200 deletes existing event	✗	✓	✓	✓	Broken due to upstream /get-event 500
	400 missing/invalid <i>eventId</i>	✗	✓	✓	✓	Missing validation
	401 unauthenticated	✗	✓	✓	✓	Missing auth check
	403 unauthorised delete by a different user	✗	✓	✓	✓	Missing ownership check
	404 event not found	✓	✓	✓	✓	

Route	Behaviour	Iter 1	Iter 2	Iter 3	Iter 4	Notes
GET /get-event	200 valid event	✓	✓	✓	✓	
	400 missing/invalid <i>eventId</i>	✗	✓	✓	✓	Missing validation
	404 event not found	✗	✓	✓	✓	Missing events returned 500 instead of 404
GET /get-all-events	200 returns events	✓	✓	✓	✓	
	200 no results	▶	✓	✓	✓	Initially untestable due to shared DB
	200 filter by source	-	✓	✓	✓	Added with filters
	200 filter by date	-	✓	✓	✓	Added with filters
	200 filter by search term	-	-	-	✓	Added with search bar
GET /get-user-made-events	200 returns events	✓	✓	✓	✓	
	200 no results	✓	✓	✓	✓	
	401 unauthenticated	✓	✓	✓	✓	
GET /:id/image	200 returns image	✓	✓	✓	✓	
	204 event exists but has no image	✗	✓	✓	✓	Originally returned 404; corrected to 204 in Iter 2
	400 missing/invalid <i>eventId</i>	-	✓	✓	✓	Added with improved event image handling
	404 event not found	✓	✓	✓	✓	

3. Iteration Evidence Summary

3.1. Authentication Iteration Summary

Iter	Change/focus	Summary	Evidence
1	Initial auth coverage	Register validation failed; login, verify, and resend passed.	Figure A.1
2	Validation fix	All original auth tests passed.	Figure A.2
3	Session persistence	Login expanded; logout and /me tests added and passed.	Figure A.3
4	Account types + code verification	Society registration and 6-digit verification added; full auth suite passed.	Figure A.4

3.2. Events Iteration Summary

Iter	Change/focus	Summary	Evidence
1	Initial event API coverage	Update, delete, and get-event tests exposed validation, auth, ownership, and not-found issues.	Figure E.1
2	Image handling + filters	Image clearing, invalid image combinations, source/date filters, and no-results behaviour passed.	Figure E.2
3	DTO refactor	Full event suite passed after standardising event fields and removing deprecated fields.	Figure E.3
4	Account-type filtering + search	Search filtering added; final event suite passed.	Figure E.4

4. Testing Limitations

The test suites focused mainly on backend API behaviour and did not provide the same level of automated coverage for frontend rendering, browser-level interaction, accessibility, or performance under load. These areas were checked partly through manual development testing, but the lack of browser-level automation means that UI regressions could still occur even where the API tests pass. Similarly, the absence of load or performance testing means the test suite validates functional correctness rather than production-level scalability.

This was an intentional trade-off, as the main technical risk during development was API contract correctness across authentication, event management, database state, image handling, and frontend/backend integration. However, future testing should add browser-level end-to-end tests, accessibility checks, and basic load testing to provide stronger evidence of reliability from the perspective of real users. The shared database also made full test isolation less straightforward, although this was mitigated through unique test data generation and targeted cleanup.

Appendix

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

> test/e2e/auth.register.test.js (3 tests | 1 failed) 337ms
  ✓ returns 201 with user payload and sends verification email on successful registration 266ms
  ✗ returns 400 when missing or invalid fields are provided 17ms
  ✓ returns 409 and does not send verification email when email already exists 52ms
✓ test/e2e/auth.login.test.js (3 tests) 387ms
✓ test/e2e/auth.verify.test.js (2 tests) 339ms
✓ test/e2e/auth.resend.test.js (2 tests) 260ms

Test Files 1 failed | 3 passed (4)
Tests 1 failed | 9 passed (10)
Start at 19:13:50
Duration 2.54s (transform 32ms, setup 0ms, import 667ms, tests 1.32s, environment 0ms)
```

[Figure A.1: Authentication Iteration 1 test run showing the failed registration validation case]

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

✓ test/e2e/auth.login.test.js (3 tests) 341ms
✓ test/e2e/auth.register.test.js (3 tests) 256ms
✓ test/e2e/auth.verify.test.js (2 tests) 274ms
✓ test/e2e/auth.resend.test.js (2 tests) 251ms

Test Files 4 passed (4)
Tests 10 passed (10)
Start at 11:36:14
Duration 2.31s (transform 29ms, setup 0ms, import 654ms, tests 1.12s, environment 0ms)
```

[Figure A.2: Authentication Iteration 2 test run showing the original authentication tests passing after validation was corrected]

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

✓ test/e2e/auth.login.test.js (4 tests) 884ms
✓ test/e2e/auth.register.test.js (3 tests) 486ms
✓ test/e2e/auth.logout.test.js (2 tests) 448ms
✓ test/e2e/auth.verify.test.js (2 tests) 439ms
✓ test/e2e/auth.me.test.js (2 tests) 452ms
✓ test/e2e/auth.resend.test.js (2 tests) 372ms

Test Files 6 passed (6)
Tests 15 passed (15)
Start at 19:07:37
Duration 4.76s (transform 34ms, setup 0ms, import 920ms, tests 3.08s, environment 0ms)
```

[Figure A.3: Authentication Iteration 3 test run showing added session persistence, logout, and /me tests passing]

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

✓ test/e2e/auth.login.test.js (4 tests) 1063ms
✓ test/e2e/auth.register.test.js (4 tests) 731ms
✓ test/e2e/auth.verify.test.js (2 tests) 634ms
✓ test/e2e/auth.me.test.js (2 tests) 604ms
✓ test/e2e/auth.logout.test.js (2 tests) 617ms
✓ test/e2e/auth.resend.test.js (2 tests) 530ms

Test Files 6 passed (6)
Tests 16 passed (16)
Start at 15:52:28
Duration 5.85s (transform 39ms, setup 0ms, import 916ms, tests 4.18s, environment 0ms)
```

[Figure A.4: Authentication Iteration 4 test run showing society registration and 6-digit verification tests passing]

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

> test/e2e/events.update-event.test.js (7 tests | 3 failed) 1943ms
  ✓ returns 200 and updates event details 499ms
  ✓ returns 200 and preserves image when updating without a new one 248ms
  ✓ returns 200 and replaces image when updating with a new one 252ms
  ✗ returns 400 when missing or invalid `eventId` is provided 179ms
  ✗ returns 401 when user is not authenticated 211ms
  ✗ returns 403 for unauthorized update by a different user 374ms
  ✓ returns 404 for non-existent event 180ms
> test/e2e/events.delete-event.test.js (5 tests | 4 failed) 1410ms
  ✗ returns 200 and deletes existing event 480ms
  ✗ returns 400 when missing or invalid `eventId` is provided 178ms
  ✗ returns 401 when user is not authenticated 205ms
  ✗ returns 403 for unauthorized delete by a different user 370ms
  ✓ returns 404 for non-existent event 175ms
> test/e2e/events.get-event.test.js (3 tests | 2 failed) 503ms
  ✓ returns 200 with event for valid `eventId` 465ms
  ✗ returns 400 when missing or invalid `eventId` is provided 18ms
  ✗ returns 404 for non-existent event 19ms
✓ test/e2e/events.get-user-made-events.test.js (3 tests) 908ms
✓ test/e2e/events.create-event.test.js (4 tests) 908ms
✓ test/e2e/events.id-image.test.js (3 tests) 688ms
✓ test/e2e/events.get-all-events.test.js (2 tests | 1 skipped) 519ms

Test Files 3 failed | 4 passed (7)
Tests 9 failed | 17 passed | 1 skipped (27)
Start at 15:06:40
Duration 8.92s (transform 49ms, setup 0ms, import 1.14s, tests 6.88s, environment 1ms)
```

[Figure E.1: Events Iteration 1 test run showing validation, authentication, ownership, and not-found failures before route fixes]

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

✓ test/e2e/events.update-event.test.js (9 tests) 2750ms
✓ test/e2e/events.delete-event.test.js (5 tests) 1616ms
✓ test/e2e/events.get-all-events.test.js (4 tests) 1320ms
✓ test/e2e/events.create-event.test.js (4 tests) 1151ms
✓ test/e2e/events.get-user-made-events.test.js (3 tests) 1070ms
✓ test/e2e/events.id-image.test.js (4 tests) 882ms
✓ test/e2e/events.get-event.test.js (3 tests) 672ms

Test Files 7 passed (7)
  Tests 32 passed (32)
Start at 13:08:12
Duration 11.47s (transform 53ms, setup 0ms, import 1.10s, tests 9.46s, environment 1ms)
```

[Figure E.2: Events Iteration 2 test run showing image handling, filtering, and no-results behaviours passing after fixes]

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

✓ test/e2e/events.update-event.test.js (9 tests) 3124ms
✓ test/e2e/events.delete-event.test.js (5 tests) 1711ms
✓ test/e2e/events.get-all-events.test.js (4 tests) 1409ms
✓ test/e2e/events.create-event.test.js (4 tests) 1188ms
✓ test/e2e/events.get-user-made-events.test.js (3 tests) 1152ms
✓ test/e2e/events.id-image.test.js (4 tests) 922ms
✓ test/e2e/events.get-event.test.js (3 tests) 692ms

Test Files 7 passed (7)
  Tests 32 passed (32)
Start at 11:34:39
Duration 12.41s (transform 55ms, setup 0ms, import 1.26s, tests 10.20s, environment 1ms)
```

[Figure E.3: Events Iteration 3 test run showing the full events suite passing after the event DTO refactor]

```
DEV v4.0.18 C:/Users/alexa/Desktop/kent-events/server

✓ test/e2e/events.update-event.test.js (9 tests) 2822ms
✓ test/e2e/events.get-all-events.test.js (5 tests) 1591ms
✓ test/e2e/events.delete-event.test.js (5 tests) 1731ms
✓ test/e2e/events.create-event.test.js (4 tests) 1168ms
✓ test/e2e/events.get-user-made-events.test.js (3 tests) 1114ms
✓ test/e2e/events.id-image.test.js (4 tests) 931ms
✓ test/e2e/events.get-event.test.js (3 tests) 755ms

Test Files 7 passed (7)
  Tests 33 passed (33)
Start at 12:29:23
Duration 12.16s (transform 54ms, setup 0ms, import 1.14s, tests 10.11s, environment 1ms)
```

[Figure E.4: Events Iteration 4 test run showing account-type filtering, search behaviour, and the final events suite passing]